

Samuel Degen

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Education	University of California Los Angeles (UCLA)	9 / 2023 - present
	B.S. Physics – GPA: 3.988 (overall); 4.0 (major); 16 graduate physics classes	
	Advisors: Zvi Bern & Mikhail Solon (Scattering Amplitudes)	
	University of Colorado Boulder	2021 - 2023
	High School Concurrent Enrollment – GPA: 4.0 (overall)	

Publications	3.	S. Wang, S. Degen and H. Liang, “Functional-Renormalization-Group Density-Matrix Functional Theory: Benchmarking the Lipkin–Meshkov–Glick model in the Normal Phase”, <i>in prep for October</i> , 2026.	[2610.XXXXX]
	2.	Z. Bern, S. Degen , E. Herrmann, M.P. Solon and A.M. Wolz, “Environmental Effects in Post-Minkowskian Dynamics: Effective Field Theory, Feynman Rules, and Ward Identities for Compact Objects in Relativistic Fluids”, <i>preprint</i> , 2026.	[2609.20742]
	1.	S. Wang, S. Degen and H. Liang, “Functional methods for quantum thermodynamics”, <i>submitted</i> , 2026.	[2605.31112]

Honors	Institutional Nomination for Marshall and Rhodes Scholarships <i>UCLA Center for Scholarships & Scholar Enrichment</i>	<i>8 / 2026</i>
	UCLA URC-Sciences Summer Program (URCSSP) <i>Undergraduate Research Center - Sciences, UCLA; Los Angeles, CA, USA</i> Advisors: Mikhail Solon & Zvi Bern (Scattering Amplitudes) Fully funded 10-week research fellowship, supporting the proposal “Precision Modeling of Environmental Effects in Gravitational Wave Signatures”	<i>Summer 2026</i>
	Goldwater Scholarship <i>Barry Goldwater Scholarship and Excellence in Education Foundation</i> - National undergraduate research scholarship recognizing exceptional promise for a research career in the natural sciences, mathematics, or engineering 1 of 454 students nationwide selected; 1 of 2 UCLA students selected in 2026	<i>2026 - 2027</i>
	APS Steven Chu Award for Best Undergraduate Research (1st) <i>2025 Annual Meeting of the Far West Section of the American Physical Society (APS); UC Santa Cruz, CA, USA</i> - Awarded 1st Prize for the talk “Towards a Nonperturbative Many-Body Nuclear Theory: Benchmarking FRG-DFT with the One-Dimensional Fermi Gas” - Selected as the top undergraduate research talk (theory or experiment) among presenters from California, Nevada, and Hawaii.	<i>10 / 2025</i>
	UCLA Undergraduate Research Scholars Program (URSP)	<i>2025 - 2026</i>

Undergraduate Research Center - Sciences, UCLA; Los Angeles, CA, USA

- 1 of 3 physics students selected for UCLA's premier undergraduate research scholarship, supporting the proposal "Effects of Black Hole Environments on Gravitational Wave Signals: A Field-Theoretic Approach" with Mikhail Solon

Mani L. Bhaumik Institute for Theoretical Physics Summer Research Fellowship

8 - 9 / 2025

Mani L. Bhaumik Institute for Theoretical Physics, UCLA; Los Angeles, CA, USA

Advisors: Zvi Bern & Mikhail Solon (Scattering Amplitudes)

- First undergraduate offered a summer research fellowship at UCLA's Bhaumik Institute for Theoretical Physics

University of Tokyo Research Internship Program (UTRIP)

6 - 7 / 2025

Quark Nuclear Science Institute, The University of Tokyo (東京大学); Tokyo, Japan

Advisor: Haozhao Liang (Nuclear Theory)

- 1 of 14 students selected from [1,149 applicants](#) for in-person, fully funded 6-week research and culture program in Tokyo
- *Friends of UTokyo, Inc. (FUTI) Global Leadership Award* – premier funding award for USA students to research in Japan (1 of 2 UTRIP students)

Perimeter Institute's PSI START Satellite Program

5 / 2025

(Perimeter Scholars International Students' Training Accelerator for Research in Theory)

Bishop's University; Sherbrooke, Quebec, Canada

- 1 of 3 international students selected for in-person, fully funded 2-week intensive theoretical physics coursework in Canada

UCLA Summer Undergraduate Research Fellowship

Summer 2024

Dept. of Physics & Astronomy, UCLA; Los Angeles, CA, USA

Advisor: E. Paulo Alves (Plasma Theory)

- 1 of 2 first-year undergraduates selected for competitive paid 10-week research fellowship

(declined) **USTC Future Scientist Exchange Program (FuSEP)**

6 - 7 / 2026

University of Science and Technology of China (中国科学技术大学); Hefei, China

- Selected for in-person, fully funded 6-week research and culture program

**Presenting
& Talks**

◇ = invited
★ = awarded

Research

"Black Hole Binary Dynamics in an Astrophysical Fluid Environment"

- EFT for Environment Project Coordinator Update, GWSky On-line Monthly Meeting *[planned March 15, 2027]*

"An Effective Field Theory Framework for Compact Objects in General-Relativistic Inviscid Fluids"

- [Summer Undergraduate Research Showcase](#), UCLA *August 26, 2026*

"Synergistic Approaches to Precision Environmental Effects in Gravitational Waveform Modeling" (with Francisco Duque)

- EFT For Environment Project Coordinators' Report, GWSky Workshop - Precision Gravitational Wave Astronomy: From Theory to Discovery, Sexten Center for Astrophysics Riccardo Giacconi; Sexten, Italy *July 29, 2026*

"Progress on Computing the Dynamical Friction Force at next order in G_N "

- GWSky Online Monthly Meeting *July 20, 2026*

- Flash Talks, GWSky Workshop - Precision Gravitational Wave Astronomy: From Theory to Discovery, Sexten Center for Astrophysics Riccardo Giacconi; Sexten, Italy *July 28, 2026*
- “Gravitational Drag from Phonon Emission”
 - Weekly Seminar, Amplitudes Journal Club, UCLA *April 23, 2026*
 - [California Amplitudes Meeting 2026 \(Davis\)](#), UC Davis *May 30, 2026*
 - Cosmological Correlators Summer School Gong Show, Institut des Hautes Études Scientifiques (IHES), France *July 10, 2026*
- “Effects of Black Hole Environments on Gravitational Wave Signals”
 - [California Amplitudes Meeting 2025 \(LA\)](#), UCLA *November 8, 2025*
 - Undergraduate Research & Creativity Showcase, UCLA *May 19, 2026*
- “Towards a Nonperturbative Many-Body Nuclear Theory: Benchmarking FRG-DFT with the One-Dimensional Fermi Gas”
 - UTRIP Final Presentation, The University of Tokyo *August 5, 2025*
 - ★ [Nuclear Physics Oral Contributions \(L04.00005\)](#), APS Far West Section Meeting, UC Santa Cruz *October 11, 2025*
 - *Steven Chu Award (1st)*
 - [Nuclear Theory Oral Contributions](#), APS Global Physics Summit 2026; Denver, Colorado *March 17, 2026*
- “Physics Without Borders: Nuclear Theory and Global Perspectives from a Summer in Tokyo”
 - ◇ Special Seminar, Physics Summer Research Experience for Undergraduates (REU), UCLA *August 15, 2025*
- “Data-driven statistical model of nonthermal particle acceleration by the kink instability in relativistic jets”
 - Department of Physics Summer Research Talks, UCLA *August 22, 2024*
 - [Plasma Astrophysics Oral Contributions \(NO05.00014\)](#), 66th Annual Meeting of the APS Division of Plasma Physics; Atlanta, Georgia *October 9, 2024*

Pedagogy

- “The Astounding Success of Effective Theories: From Quarks and Gluons to Black Hole Mergers”
 - ◇ Guest Lecture on Theoretical Physics, UCLA Olga Radko *[planned October 2026]*
Endowed Math Circle (ORMC)
- “An Introduction to the Dark Photon”
 - PHYS 226D Final Presentation, UCLA *December 4, 2025*
- “From Finite Groups to Lie Algebras: Symmetries and Representations in Physics”
 - Perimeter Institute’s PSI START Satellite Program, Bishop’s University; Quebec, Canada *May 29, 2025*
- “Convex Compactness and its Applications”
 - APPM 6560 Final Presentation, CU Boulder *May 1, 2023*

Meetings & Schools

[GWSky Workshop - Precision Gravitational Wave Astronomy: From Theory to Discovery](#)

7 / 2026

Sexten Center for Astrophysics Riccardo Giacconi, Italy

(Coordinator of the “EFT for Environments” project, organizing discussion sessions and presenting on synergistic next steps for the collaboration)

[2026 IHES Summer School - Cosmological Correlators](#)

7 / 2026

Institut des Hautes Études Scientifiques (IHES), France

(Selected participant; advanced international school primarily for graduate students and postdocs)

California Amplitudes Meeting 2026 (Davis) 5 / 2026

Center for Quantum Mathematics and Physics (QMAP), UC Davis

APS Global Physics Summit 2026 3 / 2026

Denver, Colorado

Accelerating Math and Theoretical Physics with AI 3 / 2026

Institute for Pure and Applied Mathematics (IPAM), UCLA

GWSky Kick-off Meeting 12 / 2025

Max Planck Institute for Gravitational Physics (Albert Einstein Institute-AEI), Potsdam, Germany

California Amplitudes Meeting 2025 (LA) 11 / 2025

Mani L. Bhaumik Institute for Theoretical Physics, UCLA

2025 Annual Meeting of the APS Far West Section 10 / 2025

UC Santa Cruz

2025 Southern California Strings 10 / 2025

Mani L. Bhaumik Institute for Theoretical Physics, UCLA

California Amplitudes Meeting 2025 (Davis) 5 / 2025

Center for Quantum Mathematics and Physics (QMAP), UC Davis

66th Annual Meeting of the APS Division of Plasma Physics 10 / 2024

Atlanta, Georgia

Research **UCLA – Scattering Amplitudes** 8 / 2025 - present

Experience **Advisors:** Zvi Bern & Mikhail Solon

- Developed a hydrodynamic effective field theory for compact objects in general-relativistic inviscid fluids within a unified, power-counted diagrammatic expansion by drawing connections between gravitational environmental effects and jet-quenching physics in QCD.
- Proved gauge consistency of graviton–phonon Feynman rules using generalized on-shell Ward identities in a Lorentz-broken theory, developing the framework needed for order-by-order in Newton’s G (post-Minkowskian) calculations of environmental effects on binary dynamics
- Showed that observable amplitudes in weakly self-gravitating environments factorize into vacuum-gravity and fluid sectors, allowing phonon–graviton kinetic mixing to be retained and place environmental effects under the same perturbative control as vacuum gravitational scattering calculations.
- Demonstrated power of EFT by recovering the state-of-the-art relativistic dynamical friction force for linear motion from a tree-level single-phonon emission amplitude.
- Youngest member and only undergraduate in the European Research Council’s GWSky collaboration, contributing to a new international effort on precision theory for next-generation gravitational-wave astronomy; serving as a coordinator for the *EFT for Environments* project.
- Presented this work at the GWSky Sexten Meeting, IHES Summer School, California Amplitudes Meeting, and UCLA Amplitudes Journal Club.

Supported by UCLA URCSSP, Mani L. Bhaumik Institute Summer Research Fellowship, and UCLA URSP.

Manuscript in prep for September 2026.

The University of Tokyo (東京大学) – Nuclear Theory 5 / 2025 - present

Advisor: Haozhao Liang

- Applying functional renormalization group-aided density functional theory (FRG-DFT) to exactly solvable quantum many-body systems, with the goal of benchmarking nonperturbative first-principles approaches to nuclear density functionals.
- Deriving and truncating flow-equation hierarchies for ground-state energies and correlation functions, enabling controlled comparisons between FRG-DFT, perturbation theory, and exact thermodynamics.
- Developing numerical implementations to solve coupled flow equations and test the accuracy of approximate truncations across interaction strengths and densities.
- Co-authored a manuscript on benchmarking FRG-DFT against exact thermodynamics for the single-site Bose-Hubbard model, with broader applications to strongly correlated fermions and nuclear many-body theory.
- Presented related work at the University of Tokyo, APS Far West Section Meeting, and APS Global Physics Summit 2026; awarded 1st prize of the APS Steven Chu Award for Best Undergraduate Research for this project.

Supported by UTRIP 2025 and FUTI Global Leadership Award.

Associated Publication: [arXiv:2605.31112](#)

UCLA – Plasma Theory

4 / 2024 - 1 / 2025

Advisor: E. Paulo Alves

- Developed analytic and machine-learning methods to study time-dependent nonthermal particle acceleration in relativistic astrophysical jets.
- Formulated an inverse-modeling approach to distinguish physical acceleration histories from degenerate solutions that reproduce the same observed particle spectrum.
- Demonstrated limitations of simple energy-dependent acceleration models and identified more flexible time-dependent descriptions consistent with simulated jet dynamics.
- Presented results in an oral contribution at the 66th Annual Meeting of the APS Division of Plasma Physics.

Supported by UCLA Physics Summer 2024 Research Fellowship.

UCLA – Condensed Matter

1/2024 - 3/2024

Advisor: B.C. Regan

- Developed an undergraduate-level pedagogical treatment of the Feynman checkerboard model, emphasizing its connection to propagators and relativistic quantum mechanics.

Relevant Courses	Final Papers			
	S. Degen, “ An Introduction to 3D Gravity from TQFTs ” (UCLA PHYS 242C Final Project)			6 / 2025
	S. Degen, “ Tensor Products of Representations of Lie groups ” (UCLA MATH 229A Final Project)			10 / 2024
UCLA	Graduate	PHYS 215A	Statistical Physics	
* Fall 2026		PHYS 221A	Quantum Mechanics I	
† final papers linked above		PHYS 221B	Quantum Mechanics II	
		PHYS 226B	Particle Physics (Standard Model)	
		PHYS 226C	Particle Physics (QCD and Higgs)	
		PHYS 226D	Beyond the Standard Model	
		PHYS 230A	Quantum Field Theory I	
		PHYS 230B	Quantum Field Theory II	

		PHYS 230C	Quantum Field Theory III
		PHYS 230D	Advanced Quantum Field Theory
		PHYS 231B	Mathematical Physics (Lie Theory)
		PHYS 232B	Gravitational Waves & The Double Copy
		PHYS 241A*	Solid State Physics
		PHYS 242C [†]	Topological Quantum Field Theory
		PHYS 291	String Theory Journal Club
		PHYS 495	Teaching College Physics
		MATH 229A [†]	Lie Groups and Lie Algebras
	Undergrad	PHYS 140A	Solid State
		PHYS 199*	Directed Research
		CSCI 174A	Computer Graphics (JavaScript)
		JAPAN 1-3	Elementary Modern Japanese
		CHIN 1-2	Elementary Modern Chinese
CU Boulder	Graduate	PHYS 5770	Gravitational Theory
		APPM 6560	Measure-Theoretic Probability
	Undergrad	CSCI 3104	Algorithms (C++)
		CSCI 4622	Machine Learning (Python)

Community & Service	Guest Lecturer, UCLA Olga Radko Endowed Math Circle (ORMC)		<i>2026</i>
	Invited to deliver a guest lecture on theoretical physics to 400+ gifted pre-college mathematics students in Los Angeles' largest math circle, communicating advanced ideas to a non-specialist audience while demonstrating that serious scientific research is accessible to young students		
	Peer Mentor, Division of Physical Sciences, UCLA		<i>2025 - present</i>
	Provided first-year physical science students at UCLA guidance on coursework, professional development, fellowships, and early research opportunities, especially for those interested in theoretical physics		
	Contributor, Amplitudes Journal Club, UCLA Department of Physics		<i>1 / 2025 - present</i>
	First undergraduate member and presenter of this journal club		
	Member, String Theory Journal Club, UCLA Department of Physics		<i>1 / 2025 - present</i>
	First undergraduate member of this journal club		
	Professional Events Chair, Society of Physics Students, UCLA		<i>2025</i>
	Developed extensive resources for summer research opportunities and course planning		
	Contributor, Plasma Theory Journal Club, UCLA Department of Physics		<i>4 / 2024 - 1 / 2025</i>

- Presented recent papers to theoretical plasma PhD students and faculty
- Discussed theory and application of novel analytic and machine learning methods presented in the club to current projects in the UCLA Plasma Theory group

Interviews & Press	Goldwater Scholarship (Scholars List) (UCLA College Interview) (UCLA Physics News)	<i>3 / 2026</i>
	APS Steven Chu Award for Best Undergraduate Research (APS website) (UCLA Physics News) (APS FWS Newsletter 2025 , pg. 6)	<i>10 / 2025</i>
	FUTI Global Leadership Award (report)	<i>8 / 2025</i>
	University of Tokyo Research Internship Program (program overview) (final report)	<i>6 / 2025</i>
	Perimeter Institute's PSI START Satellite (announcement) (interview)	<i>5 / 2025</i>
